

I am particularly interested in AMR applied for RTM or FWI, in geophysics. Has it been used in this area?

Yes, **AMR has been applied** to both RTM and FWI in geophysics, yielding order-of-magnitude speedups in some cases.

Has adaptive mesh refinement been used for RTM or FWI in geophysics?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

The literature confirms that adaptive mesh refinement is an active and demonstrated strategy for reducing the substantial computational cost of seismic imaging and inversion.

Adaptive Meshes in FWI

Wavefield-adapted meshes tailor element size and orientation to the expected smoothness and propagation direction of seismic waves, achieving up to an **order of magnitude** speedup in spectral-element forward and adjoint simulations (van Driel et al., 2019). This anisotropic adaptive mesh refinement (aAMR) enables discrete gradients for full waveform inversion to be computed exactly at greatly reduced cost, and the approach extends to 3D global seismic wave simulations (van Driel et al., 2019). A complete FWI workflow built on these meshes reduces computational cost by up to an order of magnitude compared to conventional approaches, provided medium property variations are sufficiently smooth (Thrustarson et al., 2019).

FWI Approach	Refinement Strategy	Key Benefit	Evidence
Wavefield-adapted (aAMR)	Anisotropic, source-centered	~10× speedup in smooth media	(Thrustarson et al., 2019; van Driel et al., 2019)
Mesh-free finite difference	Velocity-adaptive point refinement	Avoids numerical dispersion	(Takekawa & Mikada, 2016)
Triangular waveform-adapted	Element size $\propto$ local wavelength	Fewer degrees of freedom	(Roberts et al., 2021)
Optimal transport r-adaptivity	Vertex relocation to velocity field	No re-meshing I/O cost	(Santos et al., 2024)
Wavelength-adaptive stencil	G-dependent FD weights	Uniform wavefield accuracy	(Aghamiry et al., 2021)

FIGURE 1 Comparison of adaptive mesh refinement strategies for full waveform inversion

Mesh-free and meshless parameterization methods refine spatial resolution based on velocity models, dividing calculation points in low-velocity zones to provide higher resolution without complex mesh generation (Takekawa & Mikada, 2016; Peters et al., 2017). An r-adaptivity method based on optimal transport relocates mesh vertices to match a seismic velocity field, delineating sharp interfaces and salt bodies without the I/O cost of re-meshing (Santos et al., 2024). A wavelength-adaptive 27-point finite-difference stencil selects weights corresponding to local grid points per wavelength, making wavefield accuracy uniform across heterogeneous models (Aghamiry et al., 2021).

Adaptive Meshes in RTM

Velocity-adaptive unstructured meshes generated via Delaunay Triangulation and Centroidal Voronoi Tessellation use rugged topography as boundary constraints, placing coarser grids in high-velocity zones and finer grids in low-velocity zones to lower computational cost (Liu & Zhang, 2019). This approach has been applied to least-squares reverse-time migration (LSRTM), handling topography effects that conventional static corrections fail to address (Liu & Zhang, 2019).

A frequency-adaptive multiscale spatial grid uses the generalized multiscale finite-element method (GMsFEM) to solve local spectral problems on a fine grid, enabling wave simulation on coarser grids for lower frequencies and finer grids for higher frequencies (Cho & Gibson, 2018). After computing wavefields on the coarse grid, solutions are reprojected to the fine grid to construct RTM images without reducing resolution (Cho & Gibson, 2018).

Nested unstructured meshes combine low-order small meshes near the surface to accurately depict complex topography with high-order large meshes at depth for computational efficiency, with source wavefields using only the large meshes (Lu et al., 2024). Mesh-free RTM using radial-basis-function-generated finite differences reduces scattered noise from stepped approximations and avoids complex irregular grid generation (Wu et al., 2020).

Limitations and Caveats

The primary constraint of wavefield-adapted AMR is its requirement for **smooth media**, as sharp discontinuities break the approximate azimuthal symmetry of the wavefield and cause scattering that the mesh cannot capture (Thrustarson et al., 2019). In 3D global-scale applications, approximately spherical discontinuities do not pose a problem because reflected waves preserve azimuthal symmetry, but deviations from the spherically symmetric background must remain smooth (Thrustarson et al., 2019). The computational cost of additional workflow steps compared to conventional approaches is negligible, with conventional meshes serving as a fallback for complex media (Thrustarson et al., 2019). AMR has also been applied to geo-electromagnetic inversion and distributed seismic tomography in sensor networks, demonstrating broader geophysical adoption (Grayver, 2015; Kamath et al., 2016).

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

- Aghamiry, H., Gholami, A., Combe, L., & Operto, S. (2021). Accurate 3D frequency-domain seismic wave modeling with the wavelength-adaptive 27-point finite-difference stencil: a tool for full waveform inversion. *ArXiv*, abs/2108.08730. <https://doi.org/10.1190/geo2021-0606.1>
- Cho, Y., & Gibson, R. L. (2018). Reverse time migration via frequency-adaptive multiscale spatial grids. *GEOPHYSICS*. <https://doi.org/10.1190/segam2018-2988676.1>

- Grayver, A. (2015). Parallel three-dimensional magnetotelluric inversion using adaptive finite-element method. Part I: theory and synthetic study. *Geophysical Journal International*, 202, 584-603. <https://doi.org/10.1093/gji/ggv165>
- Kamath, G., Shi, L., Chow, E., & Song, W. (2016). Distributed tomography with adaptive mesh refinement in sensor networks. *Int. J. Sens. Networks*, 23, 40-52. <https://doi.org/10.1504/ijnsnet.2017.10001514>
- Liu, Q., & Zhang, J. (2019). An adaptive unstructured mesh based solution to topography least-squares reverse-time imaging. *Solid Earth Discussions*, 1-17. <https://doi.org/10.5194/se-2019-37>
- Liu, Q., & Zhang, J. (2019). Topography Least-Squares Reverse-Time Migration Based on Adaptive Unstructured Mesh. *Surveys in Geophysics*, 41, 343-361. <https://doi.org/10.1007/s10712-019-09578-0>
- Lu, Y., Lei, T., Liu, Q., Zhang, J., & Zang, N. (2024). Elastic reverse time migration based on nested triangular mesh in 2D isotropic media. *Journal of Geophysics and Engineering*. <https://doi.org/10.1093/jge/gxae058>
- Peters, F., Fontes, E. F., Mansur, W., Filho, D., Monteiro, C. S. G., & Carvalho, P. (2017). An adaptive meshless parameterization for full waveform inversion. *Engineering Analysis With Boundary Elements*, 83, 113-122. <https://doi.org/10.1016/j.enganabound.2017.07.006>
- Roberts, K. J., Olender, A., Franceschini, L., Kirby, R., Gioria, R., & Carmo, B. (2021). Full waveform inversion using triangular waveform adapted meshes. *ArXiv, abs/2108.10946*. <https://doi.org/10.48550/arxiv.2108.10946>
- Santos, T. D. D., Olender, A., Dolci, D. I., & Carmo, B. S. (2024). Velocity model-based adapted meshes using optimal transport. *GEOPHYSICS*. <https://doi.org/10.1190/geo2023-0581.1>
- Takekawa, J., & Mikada, H. (2016). A Full-Waveform Inversion with Adaptive Spatial Resolution Using a Mesh-Free Finite Difference Method. *Shigen-to-sozai*, 132, 152-159. <https://doi.org/10.2473/journalofmmij.132.152>
- Thrustarson, S., van Driel, M., Krischer, L., Boehm, C., Afanasiev, M., van Herwaarden, D. P., & Fichtner, A. (2019). Accelerating numerical wave propagation by wavefield adapted meshes. Part II: full-waveform inversion. *Geophysical Journal International*. <https://doi.org/10.31223/osf.io/v58cm>
- van Driel, M., Böhm, C., Krischer, L., & Afanasiev, M. (2019). Accelerating numerical wave propagation using wavefield adapted meshes. Part I: forward and adjoint modelling. *Geophysical Journal International*. <https://doi.org/10.31223/osf.io/43ydf>
- Wu, H., Sun, C., Li, S., Tang, J., & Xu, N. (2020). Mesh-free radial-basis-function-generated finite differences and their application in reverse time migration. *Geophysics*, 1-52. <https://doi.org/10.1190/geo2020-0015.1>